

Lok Mangal News – Full Development Roadmap (MERN Stack)

This roadmap provides a step-by-step guide for developing "Lok Mangal News" from scratch using MERN Stack (MongoDB, Express.js, React, Node.js). It covers everything from frontend, backend, database, admin panel (CMS), and deployment to ensure the website functions smoothly and dynamically updates news.

1 Project Planning & Requirements Gathering

- ✓ Define the core features of the website.
- ✓ Finalize the news categories (e.g., Breaking, Exclusive, Trending, Politics, Business, Sports, Comedy, etc.).
- ✓ Decide on the user roles (Admin for CMS management).
- ✓ Choose the tech stack:
 - Frontend: React.js (for a dynamic and interactive UI).
 - Backend: Node.js + Express.js (for API and server-side operations).
 - Database: MongoDB (to store news articles and categories).
 - CMS (Admin Panel): React-based dashboard for managing news.
 - Plan the UI/UX design (Wireframes for homepage, category pages, and admin panel).

2 Database Design (MongoDB Schema)

- ✓ Define a News Collection with fields like title, content, category, image, author, views, createdAt, isBreaking.
- ✓ Define a User Collection for admin login with username, password, role.
- ✓ Set up news categories dynamically so that new categories can be added later.

3 Backend Development (Node.js + Express.js)

- ✓ Set up Express.js server and configure MongoDB database.
- ✓ Create REST APIs to handle:
 - Adding, editing, and deleting news articles.
 - Fetching news based on categories and filters.
 - Handling user authentication (admin login).
 - Implement JWT Authentication for admin login security.
 - Enable image upload functionality for news articles.

4 Frontend Development (React.js + Redux for State Management)

- ✓ Design homepage layout with a grid to display the latest news dynamically.
- ✓ Implement a navbar & Footer with categories that filter news dynamically.
- ✓ Develop individual news article pages for detailed views.
- ✓ Create a search functionality to find news articles quickly.
- ✓ Ensure mobile responsiveness for better user experience.
- ✓ Implement auto-refresh functionality to update the latest news in real-time. (optional)

5 Admin Panel (CMS) Development

- ✓ Create a secure login system for admins.
- ✓ Develop a dashboard where admins can:
 - Add, edit, and delete news articles.
 - Upload images for news.
 - View analytics (total news posted, most viewed news, etc.).
 - Implement role-based access (only admins can modify news).
 - Provide an option to add new categories dynamically.
 - Optimize image uploads & storage to reduce server load.

6 Website Features & Functionality

- ✓ Homepage: Display breaking, trending, and latest news in a structured grid format.
- ✓ Category Pages: Each category should display relevant news dynamically.
- ✓ Auto Refresh: The homepage should fetch and display the latest news automatically. (optional)
- ✓ SEO Optimization: Implement meta tags, keywords, and Open Graph tags for better search rankings.
- ✓ Social Media Sharing: Add sharing buttons for Facebook, Twitter, WhatsApp, etc.
- ✓ Comment System : Allow users to comment on news articles.
- ✓ Newsletter Subscription (Optional): Users can subscribe to get daily news updates.

7 Testing & Debugging

- ✓ Perform unit testing on APIs and frontend components.
- ✓ Check cross-browser compatibility (Chrome, Firefox, Edge, Safari).
- ✓ Optimize page speed and performance.
- ✓ Ensure proper error handling for broken links or missing news.

8 Deployment & Hosting

- ✓ Choose a VPS (e.g., DigitalOcean, AWS, or Linode) for hosting the backend.
- ✓ Deploy the frontend on Vercel or Netlify.
- ✓ Use MongoDB Atlas for a scalable cloud database.
- ✓ Set up CI/CD pipeline for automatic updates. (optional)

9 Post-Launch Maintenance & Scaling

- ✓ Monitor server logs to track errors or crashes.
- ✓ Scale the database & backend as traffic increases.
- ✓ Implement caching (Redis) to improve load times.
- ✓ Regularly update security patches & backups.

◆ Final Deliverables

- 1 Fully functional website with news categories & filtering.
- 2 Admin panel (CMS) to manage content dynamically.
- 3 Secure authentication system for admin.
- 4 Fast & mobile-responsive UI with auto-refreshing news.
- 5 SEO-optimized website with social sharing & newsletter options.

🕒 Estimated Development Timeline

- 📌 Week 1: Project setup, database schema, and backend API development.
- 📌 Week 1-2 : Frontend UI design, navigation, and category pages.
- 📌 Week 2-3: Admin panel development and image upload integration.
- 📌 Week 3: Testing, debugging, deployment, and final optimizations.